

2023



Salesforce Migrates Its Marketing Cloud to AWS

Learn how CRM provider Salesforce is running its marketing cloud faster on AWS.

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Handles billions

49 billion daily messages handled



 2

of messages per day

during Cyber Week 2022

3–4 months to 4 hours

to deploy large-scale data tier

4–6 weeks to deploy

products on AWS

In today's hypercompetitive digital marketplace, customer demands are constantly evolving, and companies must keep up with the pace of change to stay ahead. For [Salesforce](#), one of the world's leading customer relationship management (CRM) solution providers, this means continuously innovating its Marketing Cloud solution to meet changing customer expectations. Salesforce also needed to manage growth, with the solution sending over 1 trillion messages in customer marketing campaigns. With all of Marketing Cloud's data—over 100 petabytes—stored in Salesforce's on-premises data centers, innovation was becoming expensive.

To keep up with growth while keeping costs low, Salesforce began a journey to modernize Marketing Cloud using Amazon Web Services (AWS). This initiative has involved migrating Salesforce's data tier and 200 microservices that belong to Hyperforce—a reimagination of Salesforce's platform architecture that helps organizations to grow globally and serve customers locally using cloud services such as AWS—from on-premises infrastructure to the cloud. On AWS, Marketing Cloud can scale seamlessly to deliver billions of messages per day at peak season, and Salesforce can deploy to new regions in weeks instead of months.

Opportunity | Using AWS to Scale While Cutting Infrastructure Costs for Salesforce

Founded in 1999, Salesforce provides CRM solutions to over 150,000 customers globally. Marketing Cloud, one of its largest products, offers digital marketing, engagement, communications services, and content hosting for companies from small businesses to large enterprises. By 2021, Marketing Cloud was growing 25 percent year over year, and the infrastructure became increasingly difficult and costly to manage. “Marketing Cloud has a highly distributed data tier that requires continuous innovation to continue to meet customers’ demands,” says Omar Jaber, software engineering principal architect at Salesforce. “At the same time, increased regulations, such as data residency, mean that for us to continue to grow at the pace we want to grow, we have to open data centers around the globe as well as innovate on the hardware.”

Upgrading and maintaining on-premises infrastructure involved significant time and cost investments. Salesforce thus began migrating Marketing Cloud to AWS. “After researching the possibility of running our workload in AWS, we determined that AWS was the right bet for us in terms of service availability, durability, scale, and innovation,” says Jaber. The project began in 2021, and in 14 months, Salesforce had transformed Marketing Cloud to run on AWS and deployed two new stacks in different AWS Regions. It also reduced Marketing Cloud’s infrastructure costs significantly.

Solution | Accelerating Deployments from Months to Weeks on AWS

The most critical piece of Salesforce’s solution is its massive SQL Server deployment, which comprises the heart of the data tier for Marketing Cloud. Most of the business logic for customers is stored in this SQL Server cluster, which is one of the largest in existence, and all the microservices in the solution depend on these databases. “We carry hundreds of petabytes of relational data and ran 3 trillion transactions on top of that during Cyber Week in 2021, so there’s a considerable workload on the database,” says Jaber. “The database’s success means our customers’ success.” Before migrating these workloads to AWS, the team first worked to optimize its databases. This helped the team choose the correct instance types and to design its storage solution, which uses io2 volumes and [Elastic Block Store](#) (Amazon EBS), a simple-to-use, scalable, high-performance block storage service.



Salesforce is deploying its data tier on AWS by rehosting its on-premises SQL Server databases on [Amazon Elastic Compute Cloud](#) (Amazon EC2), which provides secure and resizable compute capacity for virtually any workload. The solution has two replicas in different AWS Availability Zones for quick failover and maintains less than 10 seconds of sync between the primary and replica databases, even at peak load. Failover is less than 60 seconds, and if required, an automated process can spin a new replica on a third Availability Zone to maintain two replicas at all times. Salesforce found Availability Zones to be robust and highly available, and by deploying SQL Server using AWS, it can quickly reach new geographies that have data residency regulations. Instead of spending 1 year setting up a new data center locally, Salesforce can deploy in 4–6 weeks using AWS.

Because Salesforce stores its customer data, it makes sure to protect this data using encryption in transit and at rest. Each part of the new infrastructure on AWS uses mutual transport layer security for any communication between nodes or services. For access protocols, the solution uses [AWS Directory Service for Microsoft Active Directory](#) (AWS Managed Microsoft AD), which activates users' directory-aware workloads and AWS resources to use managed AD on AWS. Any new changes made by developers go through a design and security review before being implemented on AWS. Throughout the modernization project, Salesforce has reviewed 150 designs and performed more than 100 security assessments on the 200 microservices in its Marketing Cloud solution.

Launching new deployments on AWS is much faster than it was on premises. "One of the biggest moments for us was when we realized that deploying our large-scale data tier, which takes about 3–4 months in on-premises data centers, took only 4 hours on AWS," says Jaber. The Marketing Cloud solution can scale to meet customers' needs, and during Cyber Week 2022, the solution sent out 49 billion messages per day—including emails, notifications, WhatsApp, and text messages—for its customers' marketing campaigns. This was a 20 percent year-over-year increase compared to 2021.

Deployments for Marketing Cloud on AWS are automated, which saves time for the Salesforce team. The team doesn't have to worry about infrastructure procurement or management, either. Instead, it can focus its resources on developing innovative solutions to support customers. "Today, after our transformation, we can go to any AWS Region and build our product in about 4–6 weeks, and we are up and running, onboarding new customers," says Jaber. "This way, we can meet our growth demand and our customers' expectations."

Outcome | Continuing to Modernize Marketing Cloud Using AWS Services

Now that Salesforce has deployed Marketing Cloud on AWS, it is better equipped to meet the changing demands of the digital marketplace. It not only launches new deployments faster but also keeps them updated automatically. The company will continue to innovate rapidly—without spending a lot of time and resources on infrastructure.

"To meet the changing demand on premises, you are constantly taking time away from innovating in your own product so that you can invest in infrastructure and data tier innovation," says Jaber. "AWS provides you with the abilities and reliability that you otherwise would have to spend time and money to make happen."

About Salesforce

Salesforce is a global leader in customer relationship management technology that brings companies and customers together. With the Customer 360 platform, Salesforce connects customer data across all touchpoints, and devices.



AWS Services Used

Amazon EC2

Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute platform, with over 700 instances and choice of the latest processor, storage, networking, operating system, and purchase model to help you best match the needs of your workload.

[Learn more »](#)

Amazon EBS

Amazon Elastic Block Store (Amazon EBS) is an easy-to-use, scalable, high-performance block-storage service designed for Amazon Elastic Compute Cloud (Amazon EC2).

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AWS Directory Service for Microsoft Active Directory (AWS Managed Microsoft AD)

AWS Directory Service for Microsoft Active Directory, also known as AWS Managed Microsoft AD, activates your directory-aware workloads and AWS resources to use managed AD on AWS.

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HeyJobs on AWS

ITALY

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